



DIBYA KIRTI MISHRA

Postdoctoral Fellow
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CONTACT

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RESEARCH SKILLS

- Image Processing**
Feature identification and feature tracking using various image processing techniques in the ground and space-based data.
- Machine Learning**
Feature identification using Convolutional Neural Network (CNN) technique and other ML techniques (DBSCAN, GAN).

AREA OF RESEARCH

My research focuses on the long-term evolution of solar magnetic activity and its implications for solar dynamo theory and cycle prediction. I work extensively with century-long Ca II K and H α datasets from the Kodaikanal Solar Observatory, complemented by observations from Mount Wilson Observatory, PSPT Rome, and space-based missions including the Solar Dynamics Observatory. I specialize in developing automated algorithms (Python/IDL) for feature detection and for constructing long-term time series from historical and modern solar data. My work includes:

- Long-Term analysis of chromospheric differential rotation using the KoSO Ca II K (1904–2007).
- Automated identification of polar network bright points, enabling the reconstruction of the historical polar magnetic field back to 1904.
- Machine-learning (CNN-based) detection of plages in century-long hand-drawn suncharts.
- Construction of long-term Carrington maps combining Ca II K data.
- Multi-wavelength studies of upper-atmospheric differential rotation using SDO/AIA.
- Development of ML-based pipelines for automated detection of sunspot groups and tilt angles from KoSO white-light data.
- Participation in a project to reconstruct pre-1976 magnetograms using Ca II K and H α data through generative AI.

Overall, my expertise lies at the intersection of archival solar observations, automated feature-extraction techniques, and machine learning, aimed at improving our understanding of long-term solar variability and enhancing future solar-cycle predictions.

EDUCATION

Postdoctoral Fellow IIA - Bangalore, Karnataka (India)	2026 Sept - Now
Senior Project Associate (SPA) ARIES - Nainital, Uttarakhand (India)	2026 Jan - August
Ph. D. - Solar Astrophysics ARIES - Nainital, Uttarakhand (India)	2020 - 2025
Thesis Title - Long Term Study of Sun using Space and Ground-based Data Thesis Supervisor - Prof. Dipankar Banerjee	
M. Sc. - Physics Central University of Kerala - Kasargod, Kerala	2017 - 2019

Passed with distinction.

SKILLS

IDL	4+ yrs
Python	4+ yrs
Machine Learning	2+ yrs
LaTeX	5+ yrs

ACHIEVEMENTS

CSIR-JRF

Physics
Qualified in 2019.

DST-SERB Travel grant

Astronomy
Obtained in 2024.

PROFILES

ORCID

OTHER SKILLS

Apart from my research work, I do sketching and outreach activities.

B. Sc. - Physics
University of Delhi - Delhi

2014 - 2016

PUBLICATIONS

Differential Rotation of the Solar Chromosphere: A Century-long Perspective from Kodaikanal Solar Observatory Ca ii K Data Journal Name (ApJ) Vol 961, Issue 01, 2024, impact factor:5.4	2024
Dibya Kirti Mishra, Srinjana Routh, Bibhuti Kumar Jha, Theodosios Chatzistergos, Judhajeet Basu, Subhamoy Chatterjee, Dipankar Banerjee, Ilaria Ermolli Status: Accepted and Published	
Exploring the dynamic rotational profile of the hotter solar atmosphere: A multiwavelength approach using SDO/AIA data Journal Name (ApJ) Vol 975, Issue 02, 2024, impact factor:5.4	2024
Srinjana Routh, Bibhuti Kumar Jha, Dibya Kirti Mishra, Tom Van Doorsse-laere, Vaibhav Pant, Subhamoy Chatterjee, Dipankar Banerjee Status: Accepted and Published	
Ca II K Polar Network Index of the Sun: a Proxy for Historical Polar Magnetic Field Journal Name (ApJ) Vol 982, Issue 02, 2025, impact factor:5.4	2025
Dibya Kirti Mishra, Bibhuti Kumar Jha, Theodosios Chatzistergos, Ilaria Ermolli, Dipankar Banerjee, Lisa A Upton, M Saleem Khan Status: Accepted and Published	
Insights into Chromospheric Large-Scale Flows using Nobeyama 17 GHz Radio Observations I. The Differential Rotation Profile Journal Name (A&AL) Vol 982, 2025, impact factor:6.1	2025
Srinjana Routh, Anshu Kumari, Vaibhav Pant, Jaydeep Kandekar, Dipankar Banerjee, Mohd. Saleem Khan, and Dibya Kirti Mishra Status: Accepted and Published	
Machine Learning Based Identification of Solar Disk and Plages in Kodaikanal Solar Observatory Historical Suncharts Journal Name (ApJS) Vol 283, Issue 01, 2026, impact factor:8.5	2025
Dibya Kirti Mishra, Subhamoy Chatterjee, Bibhuti Kumar Jha, Hemapriya Raju, Aditya Priyadarshi, Manjunath Hegde, Srinjana Routh, Dipankar Banerjee, M Saleem Khan Status: Accepted and Published	

Differential Rotation of the Solar Chromosphere using multidecadal Ca II K Spectroheliograms Journal Name (IAU Proceeding), 2024	2024
Dibya Kirti Mishra, Srinjana Routh, Bibhuti Kumar Jha, Subhamoy Chatterjee, Dipankar Banerjee	

Status: Accepted and Published

Estimation of Polar Magnetic Fields using Ca II K Polar Network as a Proxy

2024

Journal Name (IAU Proceeding), 2025

Dibya Kirti Mishra, Bibhuti Kumar Jha, Theodosios Chatzistergos, Ilaria Ermolli, Dipankar Banerjee, M Saleem Khan

Status: Accepted and Published

WORKSHOPS & CONFERENCES

Space radiation workshop

ARIES, Nainital, Uttarakhand, India

24-28 Jan 2022

Online

40th Astronomical Society of India Meeting (ASI)

IIT Roorkee, Roorkee, Uttarakhand, India

25-29 Mar 2022

Presented talk titled "Study of chromospheric Differential Rotation of the Sun using Ca II K Data".

National Supercomputing Mission (NSM) workshop on High Performance Computing (HPC) for Computational Fluid Dynamics (CFD)

IIT Bombay, Bombay, India

17-20 May 2022

Participant

Aditya-L1 Science Support Cell Workshop

ARIES, Nainital, Uttarakhand, India

27 Jun-6 Jul 2022

Volunteered as project guide.

Young Astronomers' Meet (YAM), 2022

ARIES, Nainital, Uttarakhand, India

9-13 Nov 2022

Presented poster titled "Differential Rotation in the Solar Chromosphere".

Aditya-L1 Science Support Cell (AL1SSC) second workshop

Manipal Academy of Higher Education (MAHE), Manipal, Karnataka, India

28-30 Nov 2022

Participant

41st Astronomical Society of India meeting

IIT Indore, Indore, India

01-05 Mar 2023

Presented talk titled "Variation in the Chromospheric Differential Rotation over the Century".

3rd BINA Workshop, Indo-Belgian Cooperation

Graphic Era Hill University, Bhimtal, Uttarakhand, India

22-24 Mar 2023

Presented talk titled "Variation in the Chromospheric Differential Rotation over the Century".

SOLARNET Conference

8-12 May 2023

Haus H, Telegrafenberg, Potsdam, Germany

Presented talk titled "Differential Rotation of the Solar Chromosphere using multidecadal Ca K spectroheliograms" (online).

Aditya-L1 Support Cell 4th Workshop

28 Jun-7 Jul 2023

ARIES, Nainital, Uttarakhand, India

Volunteered as project guide.

International Astronomical Union Symposium (IAUS 365)

21-25 Aug 2023

Yerevan, Armenia

Presented talk titled "Differential Rotation of the Solar Chromosphere using multidecadal Ca K spectroheliograms".

seventh Aditya-L1 Support Cell workshop

21-30 May 2024

ARIES, Nainital, Uttarakhand, India

Presented talk and volunteered as project guide.

IAUG, 2024

6-15 August 2024

Cape Town, South Africa

Presented poster titled "Estimation of Polar Magnetic Fields using Ca II K Polar Network as a Proxy".

Joint Symposium of Space Climate 9 Symposium and ISEE Symposium

1-4 October 2024

Nagoya University, Nagoya, Japan

Presented talk titled "Ca II K Polar Network as A Proxy for the Estimation of Historical Polar Magnetic Field of the Sun".

Solar cycle variability: From understanding to making prediction

4-18 October 2024

ARIES, Nainital, Uttarakhand, India

Presented talk titled "Ca II K Polar Network as a Proxy to Estimate the Polar Field in the Sun".

Sun, Space Weather, and Solar-Stellar Connection

20-24 January 2025

IIA, Bengaluru, India

Presented poster titled "Automated Detection of Plages in Hand-drawn Suncharts from the Kodaikanal Solar Observatory Using Machine Learning Technique".

OTHER RESPONSIBILITIES

- Organisation and structuring of data observed from H α telescope, ARIES for the maintenance of the archival site.
- Active participation in Aditya-L1 workshops as a project mentor.

PARTIAL GUIDANCE

- **Amrutha P**, currently a MSc student at the Amrita Vishwa Vidyapeetham (Amritapuri Campus), Kollam, Kerala, India (**Summer Research Fellowship Program**).

- **Debargha Biswas**, currently a PhD student at IIST, Trivandrum, Kerala, India.
- **Rohan B. Mandrai**, currently a PhD student at IIT BHU, Varanasi, Uttar Pradesh, India.